

3.1.5 Common Built-in Functions (min, max, len, sum)

len() عناصر گنتا ہے، min() سب سے چھوٹا، max() سب سے بڑا، sum() اعداد کا مجموعہ، اور sorted() نئی ترتیب شدہ فہر...

CS XII

Chapter 3: Programming Fundamentals

Lecture 06 of 17

3.1.5 Common Built-in Functions (min, max, len, sum)

LEARN IT

Several built-in functions work across lists, tuples, sets and dictionaries. You do not call them with a dot on the object (except when a type also has its own method, such as `list.sort()`).

Function	Purpose	Works on
<code>min()</code>	Smallest element (numeric or alphabetical)	List, tuple, set, dictionary keys
<code>max()</code>	Largest element	List, tuple, set, dictionary keys
<code>len()</code>	Number of elements (for a dict: number of keys)	List, tuple, set, dictionary
<code>sum()</code>	Adds numeric elements	List, tuple, set, dictionary values via <code>sum(d.values())</code>
<code>sorted()</code>	Returns a new sorted list ; original unchanged	List, tuple, set, dictionary keys

```
s = {5, 1, 9, 3, 7}
print(len(s))
print(min(s))
print(max(s))
print(sum(s))
print(sorted(s))
```

OUTPUT

```
5
1
9
25
[1, 3, 5, 7, 9]
```

`5 + 1 + 9 + 3 + 7 = 25`. `sorted(s)` always returns a **list**, even when the input is a set.

WORKED EXAMPLE

The same five functions on four structures.

```
nums = [8, 3, 5, 3]
t = (8, 3, 5, 3)
s = {8, 3, 5}
d = {"math": 80, "eng": 70, "cs": 90}
print(len(nums), min(nums), max(nums), sum(nums), sorted(nums))
print(len(t), min(t), max(t), sum(t), sorted(t))
print(len(s), min(s), max(s), sum(s), sorted(s))
print(len(d), min(d), max(d), sum(d.values()), sorted(d))
```

OUTPUT

```
4 3 8 19 [3, 3, 5, 8]
4 3 8 19 [3, 3, 5, 8]
3 3 8 16 [3, 5, 8]
3 cs math 240 ['cs', 'eng', 'math']
```

On the dictionary, `min(d)` and `max(d)` compare **keys as strings**: "cs" comes first alphabetically, "math" last. `sum(d)` would try to add the keys "math" + "eng" + ... and fail. Use `sum(d.values())` — $80 + 70 + 90 = 240$. The set dropped the duplicate 3, so `len(s)` is 3 and `sum(s)` is 16.

sorted() versus .sort()

```
original = [8, 3, 5]
print("before sort:", original)
print("sorted():", sorted(original))
print("after sorted():", original)
original.sort()
print("after .sort():", original)
```

OUTPUT

```
before sort: [8, 3, 5]
sorted(): [3, 5, 8]
after sorted(): [8, 3, 5]
after .sort(): [3, 5, 8]
```

`.sort()` exists only on lists and rewrites the original. `sorted()` works on any iterable and leaves the original alone.

REMEMBER IT

LEN-MIN-MAX-SUM — count, smallest, largest, total. For a dictionary:

`len / min / max / sorted` use **keys**; `sum` needs `.values()`.

COMMON MISTAKE

`sum()` needs numbers. `sum(["a", "b"])` raises `TypeError`. Empty structures: `len([])` is 0, but `min([])` and `max([])` raise `ValueError` because there is no smallest of nothing.

اردو میں سمجھیں

`len()` عناصر گنتا ہے، `min()` سب سے چھوٹا، `max()` سب سے بڑا، `sum()` اعداد کا مجموعہ، اور `sorted()` نئی ترتیب شدہ فہرست دیتا ہے بغیر اصل کو بدلے ہیں۔ ڈکشنری پر `min` اور `max` کلیدوں پر چلتے ہیں؛ قدروں کا مجموعہ `sum(d.values())` سے نکالیں۔ `sort()` صرف فہرست کا میتھڈ ہے جو اصل فہرست بدل دیتا ہے۔

CHECK YOURSELF

1. For `s = {5, 1, 9, 3, 7}`, what are `len(s)`, `min(s)`, `max(s)`, `sum(s)`?
2. Does `sorted(s)` return a set or a list?
3. How do you total the values of `{"math": 80, "eng": 70, "cs": 90}`?

Answers: 1. 5, 1, 9, 25. 2. A list: `[1, 3, 5, 7, 9]`. 3. `sum(d.values())` → 240.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. `len(x)` works on:

- a. Sequences and many collections
- b. Only ints
- c. Only files
- d. Only NAND

MCQ 2. `min([3,1,2])` is:

- a. 1
- b. 3
- c. 2
- d. 6

MCQ 3. `sum((1,2,3))` is:

- a. 6
- b. 123
- c. Error always
- d. (1,2,3)

MCQ 4. `max` on a dict without a key function uses:

- a. Keys (careful!)
- b. Values always
- c. Lengths always
- d. Random

MCQ 5. These functions expect:

- a. Non-empty for min/max
- b. Always empty
- c. Only sets
- d. Only files

MCQ 6. `len("CS")` is:

- a. 2
- b. 1
- c. 0
- d. 3

Extra short questions

1. Four built-ins and a type they work on.
2. What happens `min([])`
3. `sum` of a list of strings?
4. `len` vs `count` in a list.

Extra long question

1. Given `marks = [70, 80, 90]` write expressions for `n`, `total`, `mean`, `best`, `worst` using built-ins. Show values. Warn about empty lists.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. len list; min tuple; max set; sum list of numbers.
2. ValueError.
3. TypeError — sum starts at 0.
4. len whole length; count(x) occurrences of x.

Long

1. n=len=3; sum=240; mean=80; max=90; min=70. Empty: don't divide; min/max error.
Link to Ch 4 statistics.

Next lecture: 3.1.6 Activity: Students Attendance Program